CSC 6302 - Computer Vision

Practical 6 Report

Motion, Tracking, Depth and Augmented Reality

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Target length: 4-6 pages

1. Overview

What I did: sparse LK + dense Farneback flow, my own Kalman filter through occlusion, chessboard calibration, AR cube with solvePnP, and stereo disparity. Demo video is recorded separately for submission.

2. Task 1 - Optical flow (4 marks)

LK tracks Shi-Tomasi corners with motion trails. Farneback gives dense flow; I visualise hue=direction, value=magnitude. Brightness-constancy fails when I suddenly brighten a frame - flow invents motion that is really just lighting. That is exactly the LK assumption from lectures.

Figure 1. LK trails

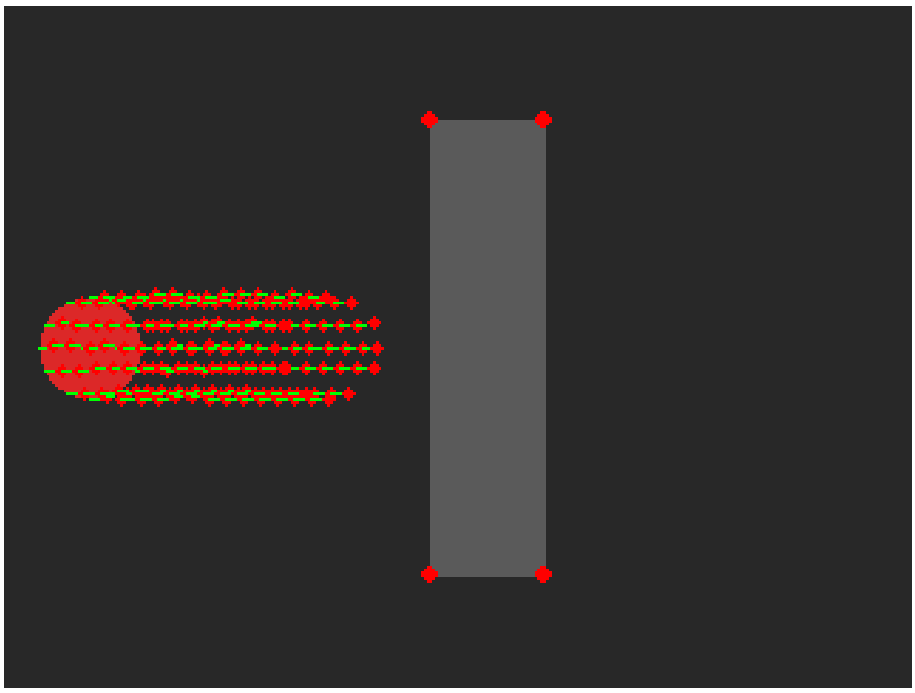


Figure 2. Farneback HSV flow

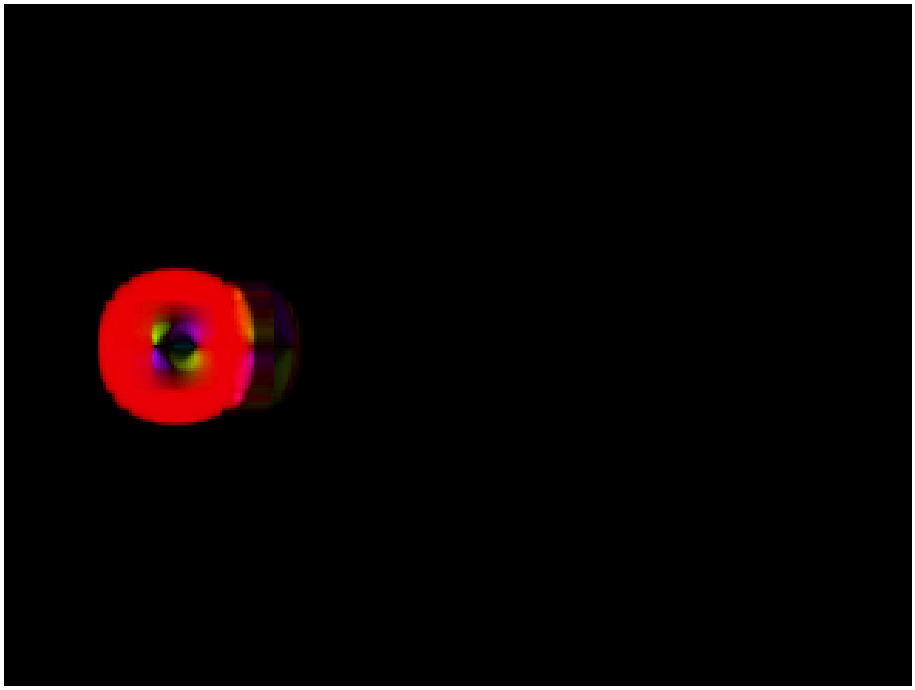
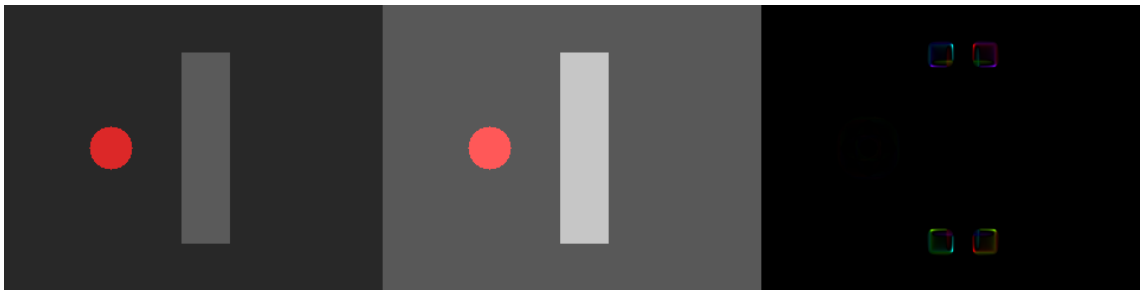


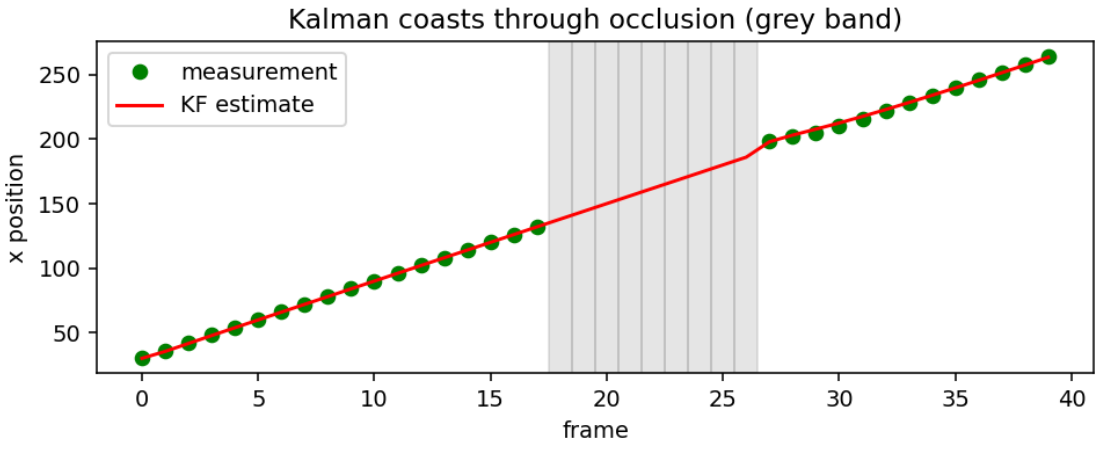
Figure 3. Flow failure under lighting change



3. Task 2 - Kalman tracking (5 marks)

What I did: HSV detect red object centroid, feed into my constant-velocity KF (predict/update written out). During occlusion z=None and the filter coasts. Tune: q=5e-2 r=8.0 (higher r = trust detector less). CSRT note: CSRT unavailable: module 'cv2' has no attribute 'TrackerCSRT_create'. CSRT/KCF often lose the lock when the object disappears; KF can keep predicting if motion is smooth.

Figure 4. Measurement vs KF (grey = occlusion)

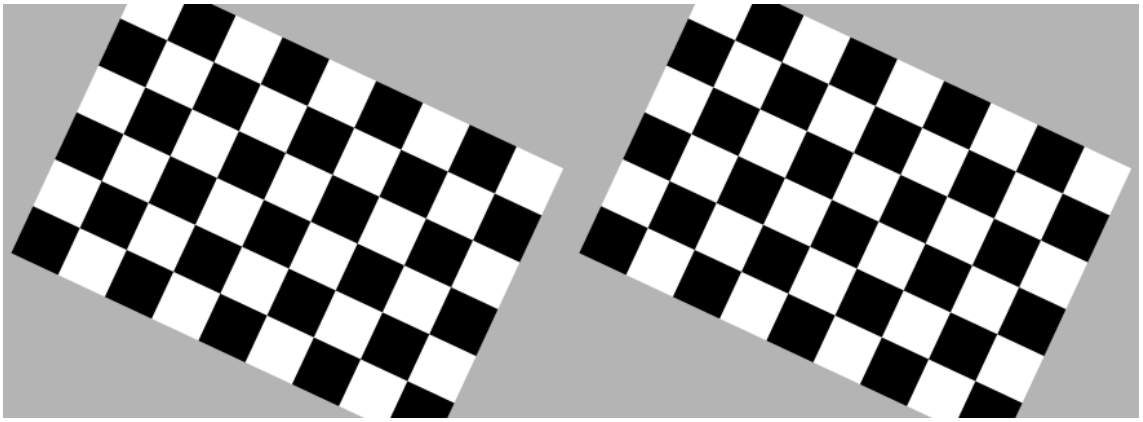


4. Task 3 - Calibration (4 marks)

What I did: findChessboardCorners + calibrateCamera on multiple views. `rms_reprojection_px=0.0371`

```
camera_matrix=
[[3.58014242e+04 0.00000000e+00 2.71365809e+02]
[0.00000000e+00 3.57988953e+04 1.28454355e+02]
[0.00000000e+00 0.00000000e+00 1.00000000e+00]]
dist_coeffs=[ 3.14012074e-01 6.00409921e-05 -3.06169676e-03 2.50142641e-03
1.57646912e-09]
fx_pixels=35801.42
views_used=11
```

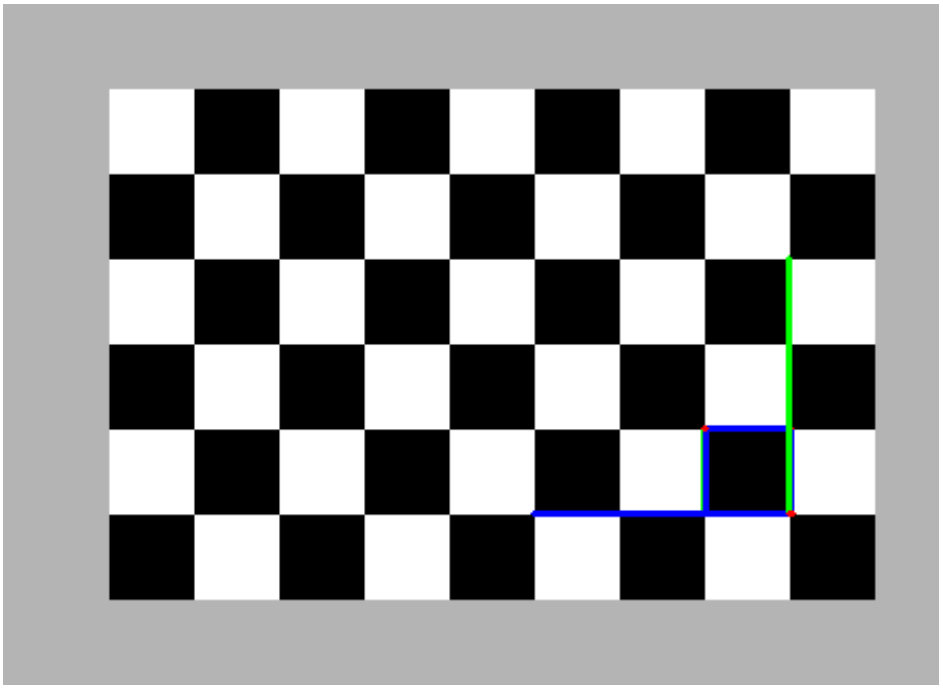
Figure 5. Before / after undistort



5. Task 4 - AR cube (4 marks)

solvePnP estimates camera pose (rvec, tvec) given known 3D board points and 2D detections plus intrinsics. Calibration had to come first because projection needs K (and distortion). I project cube vertices with projectPoints and draw edges + XYZ axes.

Figure 6. AR cube on chessboard

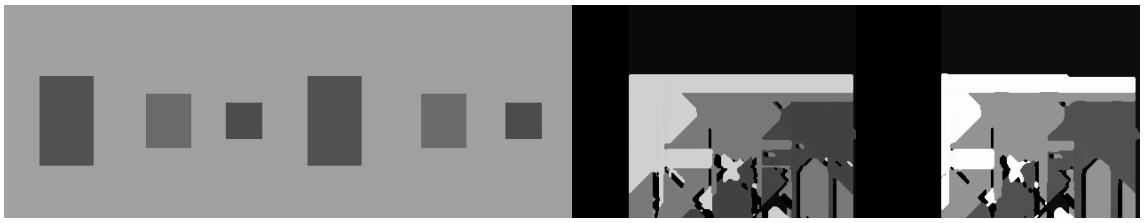


6. Task 5 - Stereo depth (2 marks)

StereoSGBM disparity; nearer should look brighter. Parameter effects shown for numDisparities/blockSize. Nearer objects should look brighter (larger disparity). Our synthetic pair is roughly rectified; real phone pairs

need proper rectification.

Figure 7. Stereo + disparity maps



References

- Gonzalez, R. C. & Woods, R. E. Digital Image Processing. (course textbook / lecture notes).
- OpenCV documentation: https://docs.opencv.org/
- Lecture slides for CSC 6302 Computer Vision (UNZA), 2025/26.
- OpenCV camera calibration and solvePnP tutorials.
- Welch & Bishop Kalman filter notes (constant velocity model).

AI use

I used **Cursor Composer** to help me get ideas for how to structure the scripts (which OpenCV functions to call, how to save figures, CLI argparse layout). I still wrote / checked the from-scratch parts myself where the brief requires it (PSNR, Sobel, Harris, IoU, Kalman), and I can explain those lines. The photos in this report are mine (or noted as demo placeholders where I say so). Any numbers in the tables come from running my code.